Green Economy & Mangroves

Strengthening Coastal Resilience & Blue Carbon

in Egypt's Red Sea · Wadi El Gemal & Hamata

Operational Implementation Strategy

Budget: €500,000 **Timeline:** 24 Months

Red Sea Coast: Wadi El Gemal · Hamata

2026–2028

The Situation on the Coast

For centuries, the Red Sea's mangrove forests have served as the ultimate coastal guardians—filtering water, sheltering juvenile marine life, and acting as formidable bio-shields against storm surges.

Today, we are facing a devastating 30% decline in mangrove cover.

This degradation is driven by rapid urbanization, unregulated camel overgrazing, and climate change. However, our inception review identified another critical issue: **Past Project Failure.** Previous international restoration initiatives often failed because they relied on guesswork—planting seedlings in the wrong sediment depths or ignoring local hydrology.

Without a rigorous, science-based intervention, the Red Sea coast faces accelerated erosion and the loss of its most productive marine hatcheries, directly threatening the livelihoods of local coastal communities.

> We must stop treating mangroves merely as 'nature to be conserved' and start treating them as the foundation of a thriving Blue Carbon Economy.

The Vision: A Blue Carbon Economy

UNESCO's vision is to replace guesswork with precision. We are shifting the paradigm from passive conservation to **Active Ecosystem Regeneration & Stewardship**.

By merging **40 years of satellite data** from the Egyptian Space Agency with the Traditional Ecological Knowledge (TEK) of the Ababda Bedouin tribes, we are building a localized "Green Economy." Here, a standing forest generates sustainable jobs in eco-tourism and fisheries, proving that conservation pays for itself.

Aligning with National & Global Mandates

This portfolio is a precisely engineered vehicle for Egypt's highest strategic commitments:

- **Egypt's NDCs (Climate Action):** Direct execution of *"Coastal Zones Adaptation"* using nature-based bio-shields against sea-level rise.
- **National Blue Economy Roadmap:** Promoting sustainable fishing, local job creation, and marine eco-tourism.
- **Climate Strategy 2050 (Goal 2b):** Mitigation of coastal erosion and extreme weather vulnerabilities.
- **UNESCO IOC Mandate:** Fulfilling the Intergovernmental Oceanographic Commission's mandate for *Ocean Literacy*.

Directly Advancing Sustainable Development Goals:

Action on the Ground: Interventions

We are moving directly into implementation across **Wadi El Gemal and Hamata** with four scientifically validated, shovel-ready activity pillars.

Precision Ecological Restoration

Target: Restore 20 km of coastline.
Action: Establishing **2 local nurseries** and planting **20,000 mangrove trees** (10k/year). Planting zones are dictated by 40-year satellite survival baselines and Ranger field logs detailing optimal salinity levels.

Protection Infrastructure

Target: Guarantee seedling survival.
Action: Installing **camel-proof fencing** to prevent overgrazing of young shoots and physically **restoring tidal channels** to ensure correct hydrological flow for roots to anchor.

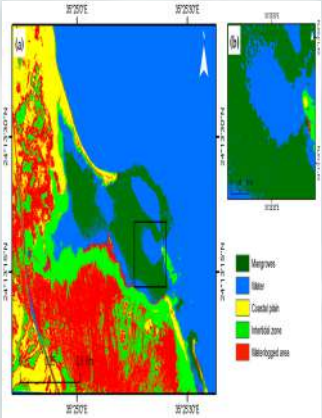
Action on the Ground: Interventions (cont.)

Green Skills TVET & Livelihoods

Target: Create sustainable "Green Jobs."
Action: Formal Vocational Training (TVET) for **60 local youth and women** in sustainable fisheries, mangrove nursery management, and eco-tourism guiding, effectively certifying locals as economic guardians.

Scientific Monitoring

Target: Validate and scale the Blue Economy.
Action: Continuous **NDVI Vegetation Index Analysis** via satellite to track forest health, culminating in a peer-reviewed *Mangrove Restoration Model* to advise national policy.



Empowering Community Stewardship

To ensure these mangroves survive beyond the 24-month funding cycle, we have institutionalized a system that marries high-tech monitoring with indigenous guardianship.

The 40-Year Satellite Baseline (EgSA)
Through our partnership with the **Egyptian Space Agency (EgSA)**, we conducted a retrospective remote sensing analysis of Earth Observation data from **1985 to 2025**. We identified exactly where mangroves historically thrived before urban encroachment, allowing us to direct planting funds with surgical precision based on 40 years of hard data.

Ocean Literacy & The Ababda Bedouin

True sustainability means local ownership. We are operationalizing the IOC's **Ocean Literacy** mandate by integrating the **Ababda Bedouin Tribes** as the central custodians of the project.
By capturing their traditional ecological knowledge alongside 20 years of unpublished Ranger planting logs, we empower the local community. Through eco-tourism and fisheries, their economic survival is now directly tied to the health of the mangrove bio-shields.

Partners in Blue Carbon

We have moved beyond informal agreements to establish a robust governance mechanism: the **Community Mangrove Stewardship Committee (CMSC)**. This unified coalition links executive power, scientific validity, and local enforcement.

1. Governance & Site Authority

Providing legal frameworks, protected area enforcement, and tourism strategy.

- **Executive:** Governorate of the Red Sea & Ministry of Environment (EEAA).
- **Site Management: Wadi El Gemal National Park Authority** (Rangers).
- **Economy:** Ministry of Tourism & Antiquities (MOTA) & Local Tourism Chamber.

2. Technical & Scientific Validation

Providing data verification, carbon analysis, and vocational certification.

- **Monitoring Lead: Egyptian Space Agency (EgSA)** (Remote Sensing).
- **Research Partners: AUC (CARES)** & **NIOF** (Oceanography Institute).
- **Training:** Local Vocational Training Centers (TVET).

3. Community & Civil Society

Executing the fieldwork, advocacy, and long-term ecosystem guardianship.

- **Local Custodians: Ababda Bedouin Tribes**.
- **NGOs: HEPCA**, **Marine Bois**, and **Mangrove Bees (NISGP)**.

Together, we are engineering a future where the ocean sustains the people.

In September 2025, King Felipe VI and President El-Sisi signed the ***Spain–Egypt Alliance for Sustainable Development*** **(2025–2030)**. This project is its direct, high-visibility operational expression.

Economic Transition
Bilateral trade nearing €3bn · Fostering high-productivity Green & Blue Economy jobs

Ecological Transition
Nature-based climate adaptation · Coastal bio-shields — *this project is the proof of concept.*

Social Transition
Women's economic empowerment · TVET skills · Protection of indigenous heritage & territory

Ready for Implementation.

Ahmed Sakna, Project Specialist · a.sakna@unesco.org

UNESCO Regional Office for Egypt and Sudan